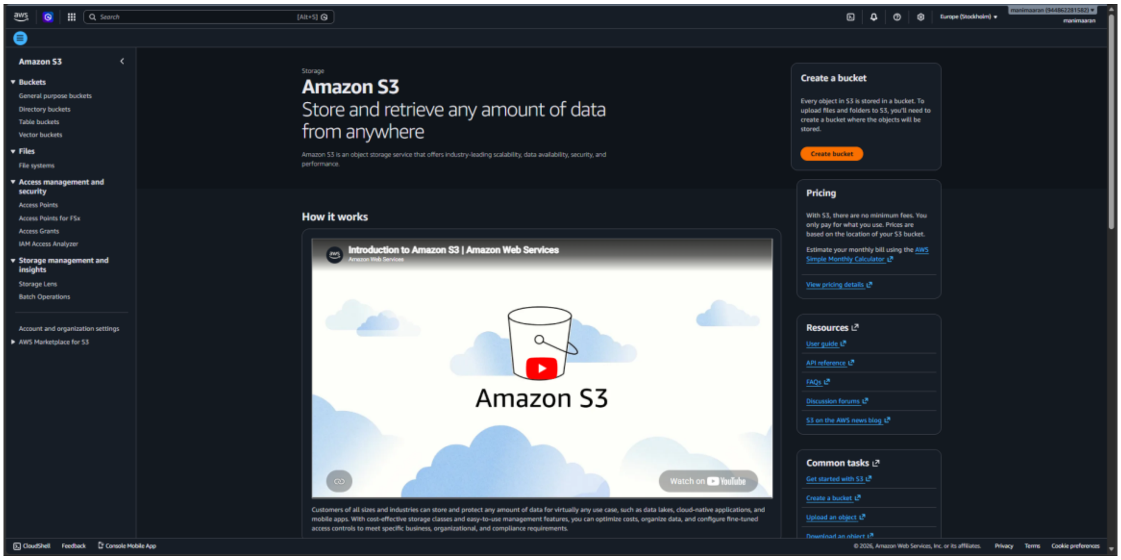
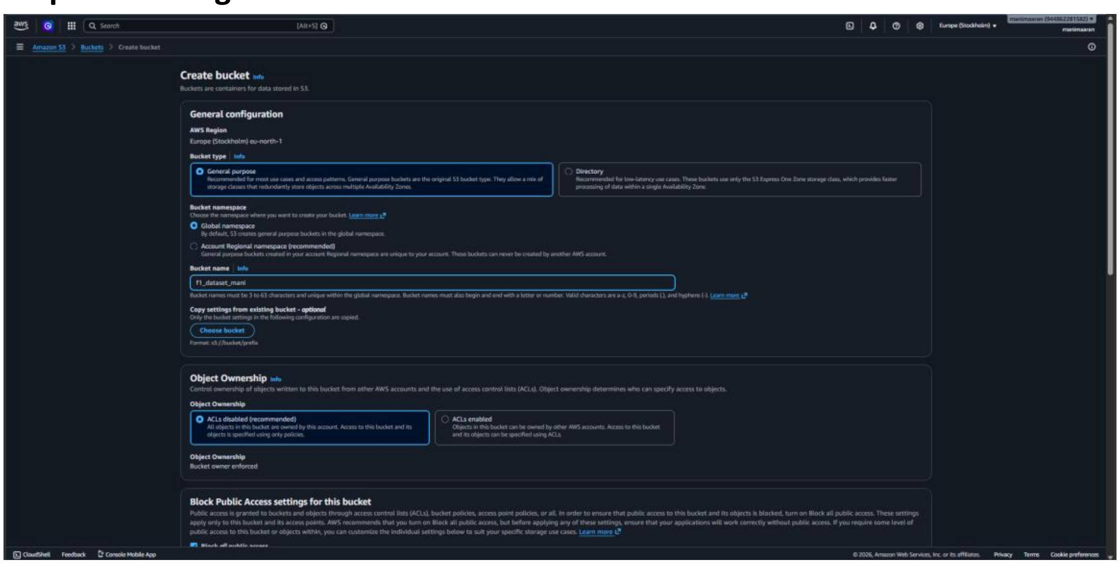
AWS S3 Setup for Databricks and Snowflake Project

Step 1: Amazon S3 Landing Page



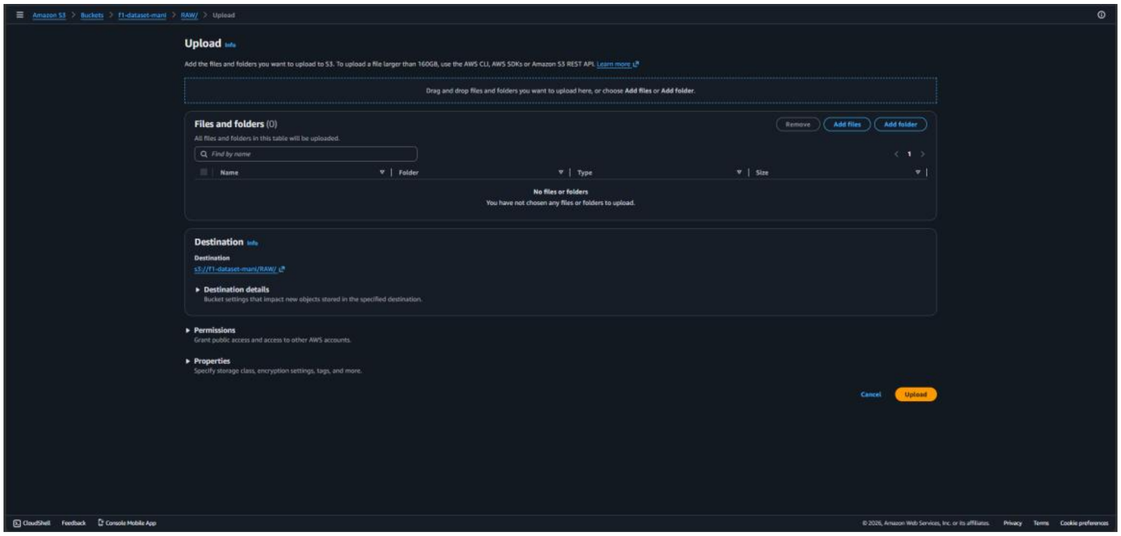
The AWS Management Console was opened and Amazon S3 was selected. Amazon S3 serves as the cloud storage layer for the Databricks and Snowflake project. It is used to store raw data files before they are processed and loaded into analytics platforms.

Step 2: Creating the S3 Bucket



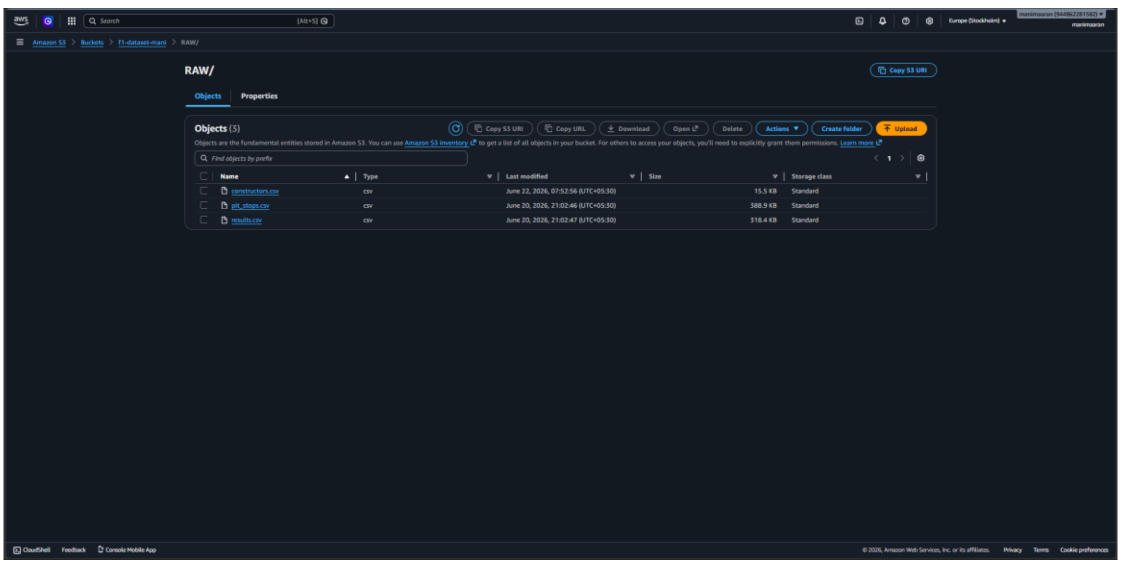
A new S3 bucket named 'f1_dataset_mani' was created in the Europe (Stockholm) region. The bucket type was set to General Purpose with default security settings. This bucket acts as the centralized storage repository for the project datasets.

Step 3: Uploading Data Files



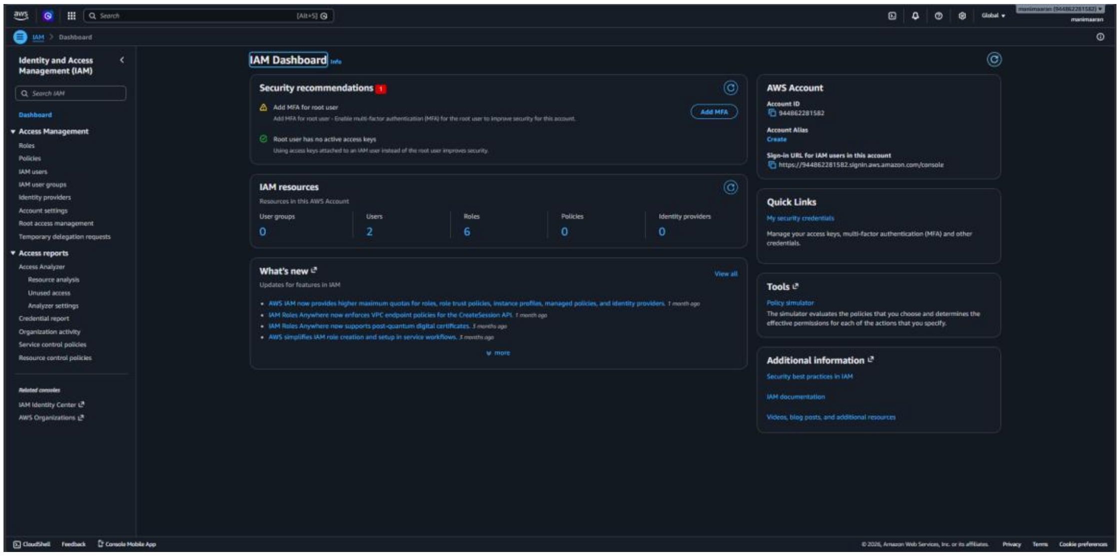
The upload interface was used to add project datasets into the bucket. The RAW folder was selected as the destination location. This step prepares the source data that will later be accessed by Databricks and Snowflake.

Step 4: Verifying Uploaded Files



After the upload process completed, the RAW folder contained the project files: constructors.csv, pit_stops.csv, and results.csv. These files form the source dataset for data engineering and analytics workflows.

Step 5: IAM Configuration



AWS IAM was accessed to manage permissions and security. IAM users, roles, and policies are required so that Databricks and Snowflake can securely access the S3 bucket without exposing root account credentials. This is a critical step for establishing secure data integration.

Integration Summary

The S3 bucket was created as the landing zone for Formula 1 datasets. Data files were uploaded into the RAW folder and validated. IAM resources were configured to support secure access. Databricks can ingest these files from S3 for transformation and processing, while Snowflake can consume the processed data through external stages or data loading pipelines. This architecture enables a scalable and secure cloud-based analytics workflow.